# Lab-4: VibeCoding

ChatGPT Interaction History and Implementation Log

**Student:** Gowthamm08 **Project:** 44_simple-platformer **Repository:** github.com/Gowthamm08/44_simple-platformer

Purpose: This document records the main steps and assistance used while completing Lab-4. It is a compiled interaction log for submission, not a verbatim export of every message in the original ChatGPT conversation.

## 1. Assignment and repository

**User request:** The student shared the Lab-4 instructions and the assigned repository SETAPESU26/44_simple-platformer.

**Task requirements identified:** fix platform collision detection; add a Game Over screen with final score; add replay with Easy, Medium, and Hard difficulty; add sound effects for jumping, reaching the goal, and dying. Deliverables: before/after gameplay videos, updated code, and the AI chat history exported as a document or PDF.

## 2. Environment setup

**Git installation:** Git was initially unavailable in Command Prompt. The student installed Git for Windows and verified it with `git --version`.

**Clone repository:** The student forked the assigned repository to their account and cloned their personal fork.

**Python:** The terminal showed Python 3.11.9.

**Dependencies:** The student ran `pip install -r requirements.txt`; Pygame installed successfully.

**Before video:** The student reported recording the original gameplay before changing the code.

## 3. Repository and code changes

The student edited `main.py` and `game/game_engine.py` in VS Code. The proposed engine update included swept vertical landing detection, a Game Over overlay, a final-score display, difficulty selection and replay, and generated sound effects using Pygame. A `.gitignore` was also created to exclude Python cache files and virtual environments.

**Important testing note:** The code was pushed successfully, but the student should verify the updated game behavior before relying on the after video. The final submission should contain only files that have been tested.

## 4. Git commands used

```
git clone https://github.com/Gowthamm08/44_simple-platformer.git
cd 44_simple-platformer
python --version
pip install -r requirements.txt
python main.py

# Ignore generated Python files
__pycache__/
*.py[cod]
.venv/

git add main.py game/game_engine.py .gitignore
git config --global user.name "Gowthamm08"
git config --global user.email "YOUR_GITHUB_EMAIL"
git commit -m "Complete Lab 4 platformer improvements"
git push origin main
```

Git initially required a configured author name and email. After setting these, the commit succeeded. The terminal confirmed the push to the personal GitHub repository, with commit `cf7e34c` and the message `Complete Lab 4 platformer improvements`.

## 5. Submission folder

```
Lab-4/
■■■ before.mp4
■■■ after.mp4
■■■ Lab-4-ChatGPT-History.pdf
```

```
■■■ updated-code/  (if required by the instructor)
    ■■■ main.py
    ■■■ game/
        ■■■ game_engine.py
        ■■■ player.py
        ■■■ platform.py
        ■■■ hazard.py
```

The student created a `Lab-4` folder in the personal repository and added a README. The before and after videos and this PDF still need to be uploaded there if they have not already been added.

## 6. Main user prompts and assistant support

**User:** Shared the lab instructions and assigned GitHub repository; asked for guidance from the beginning.

**Assistant:** Explained forking, installing Git, cloning the repository, installing requirements, and running the game.

**User:** Shared terminal errors where Git was not recognized.

**Assistant:** Guided installation of Git for Windows and verification using git --version.

**User:** Shared successful clone and Python version output.

**Assistant:** Guided installing Pygame and running main.py.

**User:** Asked to debug the code in their personal repository and said the before video was recorded.

**Assistant:** Provided replacement code for main.py and game/game_engine.py covering collision handling, Game Over, replay/difficulty, and sounds.

**User:** Shared Git status and commit errors.

**Assistant:** Explained staging, .gitignore, configuring Git identity, committing, and pushing.

**User:** Shared successful push output and asked about uploading videos.

**Assistant:** Explained creating a Lab-4 folder and uploading videos and the chat-history PDF.

## 7. Final checklist

■ Confirm the updated game runs and all four required features work.

■ Keep the original before video and the after video.

■ Upload both videos to the Lab-4 folder in the lab submission repository.

■ Upload this PDF as the AI interaction history.

■ Ensure the updated code is available in the personal GitHub repository.